

Chapter 1

Introduction: Databases and Database Users

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Outline

- Basic Definitions
- Example of a Database (UNIVERSITY)
- Typical DBMS Functionality
- Main Characteristics of the Database Approach
- Database Personnel
- When Not to Use Databases

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Basic Definitions

- **Database:**
 - A collection of related data.
- **Database Management System (DBMS):**
 - A software package that facilitates the creation and maintenance of a database.
- **Database System:**
 - The DBMS software together with the database. Sometimes, the applications (developed to use the existing database) are also included.

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Simplified database system environment

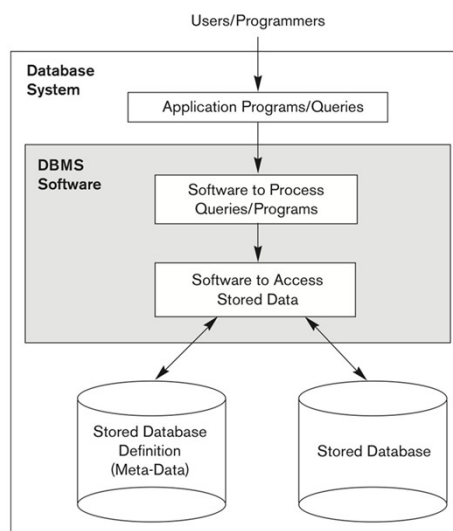


Figure 1.1 A simplified database system environment.

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Example of a simple database

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2 A database that stores student and course information.

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Typical DBMS Functionality

- *Define a particular database* in terms of its data structures, data types and constraints.

CREATE TABLE STUDENT

```
( Name VARCHAR(25) NOT NULL,
  Student_number INT NOT NULL,
  Class INT,
  Major VARCHAR(10),
  PRIMARY KEY (Student_number) );
```

- *Construct (or load) the initial database contents* on a secondary storage medium.
- *Manipulating the database:*
 - Retrieval: Querying, generating reports
 - Modification: Insertions, deletions and updates to its content on the database
 - *Concurrent processing of user queries and application programs*, while keeping all data valid and consistent.

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Example of a Database

- **Mini-world:**
Some part of the real world about which data is stored in a database.
- **Mini-world for the example database:**
Part of a UNIVERSITY environment.
- **Some mini-world *entities*:**
 - STUDENTs
 - COURSEs
 - SECTIONs (of COURSEs)
 - (academic) DEPARTMENTs
 - INSTRUCTORs

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Example of a Database (cont'd)

- **Some mini-world *relationships*:**
 - SECTIONs *are of specific* COURSEs
 - STUDENTs *take* SECTIONs
 - some COURSEs *have prerequisite* COURSEs
 - INSTRUCTORs *teach* SECTIONs
 - COURSEs *are offered by* DEPARTMENTs
 - STUDENTs *major in* DEPARTMENTs

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DB Approach vs. the Traditional File Processing Approach

- In traditional file processing, data definition is typically a part of the application programs. Hence, these programs are constrained to work with data files whose structures are defined in them.
- In the DB approach, data file structures and indexes can be changed without affecting application programs.

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Main Characteristics of the Database Approach

- **Self-describing (or self-contained) nature of a database system:**
 - The DB system contains not only the DB itself but also a complete description of the DB.
 - The description of the DB is called **meta-data**.
 - A **DB system catalog** stores the meta-data of a particular database (e.g., data structures, data types, and constraints).
- **Insulation between programs and data:**
 - Called **program-data independence**.
 - Allows changing data structures and storage organization (i.e., file structures and indexes) without having to change the DB access programs.

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Example of a Simplified Database Catalog

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

Figure 1.3 An example of a database catalog for the database in Figure 1.2.

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Main Characteristics of the Database Approach (cont'd)

- **Data Abstraction:**
 - A **data model** is used to present the users with a **conceptual representation** of the database.
 - Application programs refer to the **data model constructs (aka logical concepts)** rather than data storage details. Hence, the **data model hides storage details**.
- In other words, a data model is used to represent the entities and their relationship, and user queries can refer to the data model, without specifying storage details (i.e. file structures and indexes). 12

Example of a Relational Data Model

STUDENT

Name	Student_number	Class	Major
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Figure 2.1 Schema diagram for the database in Figure 1.2.

COURSE

Course_name	Course_number	Credit_hours	Department
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PREREQUISITE

Course_number	Prerequisite_number
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SECTION

Section_identifier	Course_number	Semester	Year	Instructor
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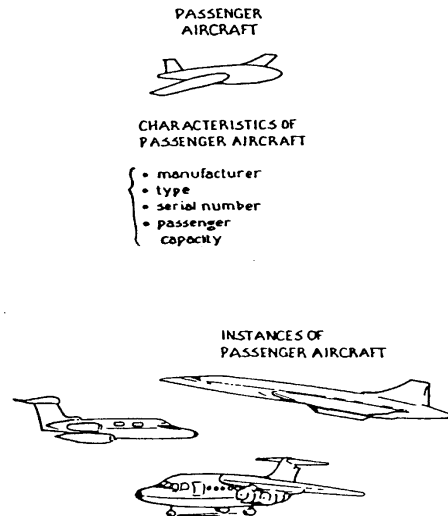
GRADE_REPORT

Student_number	Section_identifier	Grade
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Object-Oriented Data Modeling

- Object type (or class) is an abstraction of a set of real-world things (objects) such that:
 - All the objects in the set have the same characteristics.
 - All the objects in the set are subject to and confirm to the same rules.

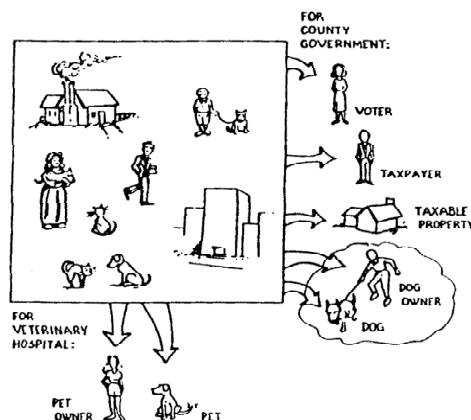
OO Data Modeling (cont'd)



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OO Data Modeling (cont'd)

- Identifying object types depends on the purpose of modeling.



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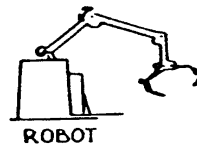
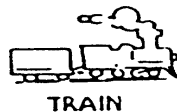
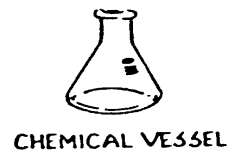
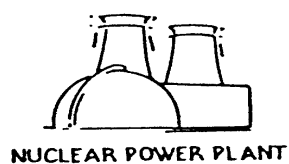
OO Data Modeling (cont'd)

- Identifying object types: object types can be identified from
 - Tangible things
 - Roles
 - Incidents
 - Interactions
 - Specifications

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OO Data Modeling (cont'd)

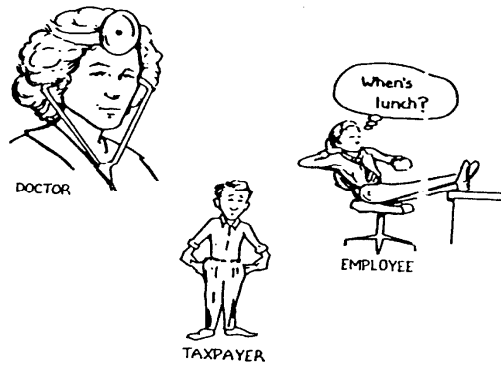
- **Tangible things.**



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OO Data Modeling (cont'd)

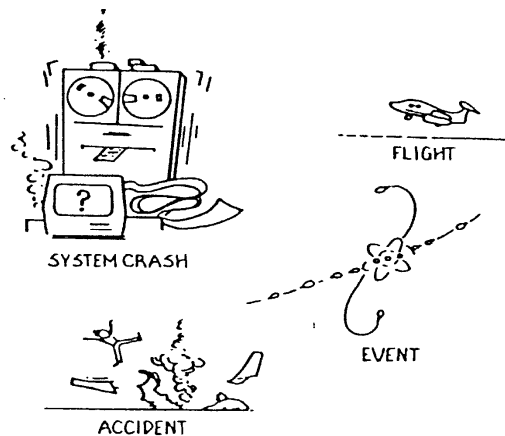
- **Roles** played by persons or organizations.



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OO Data Modeling (cont'd)

- **Incidents**: occurrences or events.

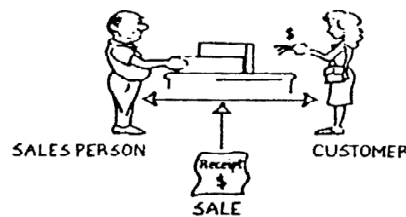


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OO Data Modeling (cont'd)

- **Interactions:** interaction object types usually have a transaction or contract quality, and related to two or more other objects.

*purchase (related to buyer, seller, and thing purchased)
marriage (related to man and woman)*



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OO Data Modeling (cont'd)

- **Specification:** specification object types frequently show up in inventory or manufacturing applications.
 - Specification of product models or products.

*The specification:
What it means to be a
Model 172*

REFRIGERATOR MODEL

Model #	Cubic feet	Power rating	...
169A	10		
172	12		

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Main Characteristics of the Database Approach (cont'd)

- **Support of multiple views of the data:**
 - Each user may see a different view of the database, which describes only the data of interest to that user.

TRANSCRIPT

Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

(a)

COURSE_PREREQUISITES

Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

(b)

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View

- In relational database systems, **a view is a named, derived, virtual table** defined in SQL.
 - A **base table** is a persistent data table (i.e., a **relation**).
- Definition of COURSE_PREREQUISITES in SQL:

```

CREATE VIEW COURSE_PREREQUISITES
    (Course_name, Course_number, Prerequisites)
AS SELECT Course_name, Course_number,
    Prerequisite_number
FROM COURSE, PREREQUISITE
WHERE Course_number=Course_number;
  
```

Main Characteristics of the Database Approach (cont'd)

- **Controlling redundancy in data to prohibit inconsistencies among the files.**

Figure 1.6

Redundant storage of Student_name and Course_name in GRADE_REPORT.
(a) Consistent data. (b) Inconsistent record.

GRADE_REPORT

Student_number	Student_name	Section_identifier	Course_number	Grade
17	Smith	112	MATH2410	B
17	Smith	119	CS1310	C
8	Brown	85	MATH2410	A
8	Brown	92	CS1310	A
8	Brown	102	CS3320	B
8	Brown	135	CS3380	A

(a)

GRADE_REPORT

Student_number	Student_name	Section_identifier	Course_number	Grade
17	Brown	112	MATH2410	B

(b)

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Main Characteristics of the Database Approach (cont'd)

- **Sharing of data and multi-user transaction processing:**
 - Allowing a set of concurrent users to retrieve/update from/to the database.
 - **Concurrency control** software within the DBMS guarantees that each transaction is correctly executed or aborted, even though multiple transactions are executed concurrently.
 - A transaction is an execution of a user program that retrieves and/or changes the content of a database.
 - **Recovery** subsystem ensures the recovery of the database from the system failure in the middle of transactions.

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Main Characteristics of the Database Approach (cont'd)

- **Restricting unauthorized accesses**
 - **Access privileges can be assigned at**
 - **Account level**
 - **Relation (i.e., table) level**

Examples:

```
GRANT CREATETAB TO Account1;  
GRANT INSERT, DELETE ON STUDENT TO Account2;  
GRANT SELECT ON STUDENT TO Account2 WITH GRANT OPTION;  
GRANT UPDATE ON STUDENT(Class) TO Account3;  
REVOKE UPDATE ON STUDENT(Class) FROM Account3;
```

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Main Characteristics of the Database Approach (cont'd)

- **Providing multiple interfaces to different classes of users:** query languages, programming language interfaces, menu-driven interfaces, natural language interfaces, etc.
- **Representing complex relationships among data.**
- **Enforcing integrity constraints on the database.**
- **Providing backup and recovery services.**

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Database Personnel

- **User Group**
 - Database Administrator (DBA)
 - DB designer
 - End users
 - System analysts and application programmers
- **System Group**
 - DBMS designers and developers
 - Tool developers
 - Operators

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Database Users

- **Database Administrator (DBA):**

Responsible for authorizing access to the database, coordinating and monitoring its use, acquiring software and hardware resources, controlling its use and monitoring the efficiency of operations.
- **Database Designers:**

Responsible to define the structure, content, and constraints of the DB. They must communicate with the end-users to understand their needs.

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Database Users (cont'd)

- **End-users**

They use the data for queries, reports, and (some of them) update the database content.

- **System Analysts and Application Programmers**

- System analysts determine the operation requirements of end users, and develop specifications for transactions (i.e., input, output and behavior of transactions).
- Application programmers implement these specifications as programs; then test, debug, document and maintain the programs.

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When not to use a DBMS

- Main inhibitors (i.e., costs) of using a DBMS:
 - High initial investment and possible need for additional hardware and administrators.
 - Run-time overhead for providing security, concurrency control, recovery, and enforcing constraints.
- When a DBMS may be unnecessary:
 - If the database and applications are simple, well defined, and not expected to change (in structure and constraints).
 - If there are stringent real-time requirements that may not be met because of DBMS overhead.
 - If access to data by multiple users is not required.

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